Math 67 – HW 2 Solutions

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Chapter 3

C2. Given any complex number $\alpha \in \mathbb{C}$, consider the polynomial

$$\begin{aligned}(z-\alpha)(z-\overline{\alpha}) &= z^2 - \alpha z - \overline{\alpha} z - \alpha\overline{\alpha}\\ &= z^2 - (\alpha+\overline{\alpha})z - |\alpha|^2.\end{aligned}$$

Its coefficients 1, $(\alpha+\overline{\alpha}) = 2\text{Re }\alpha$, and $|\alpha|^2$ are all real.

P2. Given a polynomial $p(z) = \sum_{j=0}^n a_j z^j = a_0 + a_1 z^1 + \cdots + a_n z^n$ with complex coefficients, the conjugate of $p(z)$ is defined as $\overline{p}(z) = \sum_{j=0}^n \overline{a_j} z^j = \overline{a_0} + \overline{a_1} z^1 + \cdots + \overline{a_n} z^n$.

(a) $\overline{p(z)} = \overline{p}(\overline{z})$

Proof. Using the properties of conjugation given in Theorem 2.2.9,

$$\overline{p(z)} = \overline{\left(\sum_{j=0}^n a_j z^j\right)} = \sum_{j=0}^n \overline{a_j z^j} = \sum_{j=0}^n (\overline{a_j})(\overline{z^j}) = \sum_{j=0}^n \overline{a_j}(\overline{z})^j = \overline{p}(\overline{z})$$

□

(b) $p(z)$ has real coefficients if and only if $\overline{p}(z) = p(z)$.
(Note that since this is an *if and only if* statement, we have to prove both directions.)

Proof. ($\Rightarrow$): Suppose $a_0, a_1, ..., a_n \in \mathbb{R}$ (the coefficients are real numbers). Then $\overline{a_0} = a_0, \overline{a_1} = a_1, \ldots, \overline{a_n} = a_n$, so it follows that

$$\overline{p}(z) = \sum_{j=0}^n \overline{a_j} z^j = \sum_{j=0}^n a_j z^j = p(z). \tag{1}$$

($\Leftarrow$): (Now we prove the other direction.) Suppose (1) is true. Then term by term we have $\overline{a_0} = a_0, \overline{a_1} z = a_1 z, \ldots, \overline{a_n} z^n = \overline{a_n} z^n$ for any $z \in \mathbb{C}$, so $\overline{a_0} = a_0, \overline{a_1} = a_1, \ldots, \overline{a_n} = a_n$. Thus, all of the coefficients $a_0, a_1, \ldots, a_n$ are real. □

(c) Given polynomials $p(z)$, $q(z)$, and $r(z)$ such that $p(z) = q(z)r(z)$, it follows that $\overline{p}(z) = \overline{q}(z)\overline{r}(z)$.

Proof. We are given that $p(z) = q(z)r(z)$ for any $z \in \mathbb{C}$. Then $p(\overline{z}) = q(\overline{z})r(\overline{z})$. Conjugating both sides and applying Theorem 2.2.9 yields

$$\overline{p(\overline{z})} = \overline{q(\overline{z})r(\overline{z})} = \overline{q(\overline{z})} \cdot \overline{r(\overline{z})}.$$

Finally, by the result of part (a) we have $\overline{p}(z) = \overline{q}(z)\overline{r}(z)$. □

Chapter 4

C1. (a) yes, (b) yes, (c) No. no zero element, (d) No. the set contains $(1,0)$, but not its additive inverse, which is $(-1,0)$, (f) yes, (g) No. no zero.

C3. Subspace of $\mathcal{C}(\mathbb{R})$?

(a) $U = \{ f \in \mathcal{C}(\mathbb{R}) \mid f(x) \le 0, \forall x \in \mathbb{R} \}$ **is not a subspace**. The constant function $f(x) = -1 \in U$ but its additive inverse, the function $f(x) = 1 \notin U$.

(b) $U = \{f \in \mathcal{C}(\mathbb{R}) | f(0) = 0\}$ **is a subspace**. The zero function $0 \in U$. For any two functions $f, g \in U$ we have $f(0) = 0$ and $g(0) = 0$, so their sum evaluated at zero is $(f+g)(0) = f(0) + g(0) = 0$. Hence, the function $(f+g) \in U$ and thus, U is closed under addition. The set U is closed under scalar multiplication too: observe that for function $f \in U$ and real scalar $\alpha \in \mathbb{R}$, $(\alpha f)(0) = \alpha f(0) = 0$, so the function $\alpha f \in U$.

(c) $U = \{f \in \mathcal{C}(\mathbb{R}) | f(0) = 2\}$. **Not a subspace**. U does not contain the zero function.

(d) The set of all constant functions **is a subspace**. The zero function $f(x) = 0$ (for all x) is a constant function (this is often written as $f \equiv 0$). If we add any two constant functions, their sum is a constant function, so the set of constant functions is closed under addition. If we multiply a scalar times a constant function, the resulting function is also constant, so constant functions have closure under scalar multiplication.

C5. $\mathbb{F}[z]$ is the subspace of polynomials with coefficients in $\mathbb{F}$ and $U = \{ az^2 + bz^5 \mid a, b \in \mathbb{F} \}$ is a subspace of $\mathbb{F}[z]$. Let W be the set of polynomials that have no z^2 or z^5 term, i.e.

$$\begin{aligned} W &= \left\{ \sum_{j=1}^n a_j z^j \;\middle|\; n \in \mathbb{Z}_+, a_j \in \mathbb{F}, a_2 = a_5 = 0 \right\}\\ &= \left\{ \sum_{j=1}^n a_j z^j \in \mathbb{F}[z] \;\middle|\; a_2 = a_5 = 0 \right\}\\ &= \{ p \in \mathbb{F}[z] \mid p \text{ has no 2nd or 5th order terms} \}. \end{aligned}$$

Note equivilent ways to denote the same set. We have $\mathbb{F}[z] = U + W$ because any polynomial $p \in \mathbb{F}[z]$ can be expressed as the sum of a polynomial with z^2 and z^5 terms, and one with terms of other order. Since $U \cap W = \{0\}$ (the only element that U and W have in common is the zero function $p(z) = 0$), we have by proposition 4.4.7 that $\mathbb{F}[z] = U \oplus W$ (the sum is a *direct* sum, which is a stronger statement).

P2. Let V be a vector space over $\mathbb{F}$ and let W_1 and W_2 be subspaces of V. Then their intersection $W_1 \cap W_2$ is also a subspace of V.

Proof. (zero element): Certainly since $0 \in W_1 \subseteq V$ and $0 \in W_2 \subseteq V$, it follows that $0 \in W_1 \cap W_2$.

(closure under addition): Suppose $x, y \in W_1 \cap W_2$. Then $x \in W_1$ and $y \in W_1$. Since W_1 is a subspace (which is closed under addition), $x + y \in W_1$. Similarly, since $x, y \in W_2$ and W_2 is a subspace it follows that $x + y \in W_2$. Then $x + y \in W_1 \cap W_2$.
(closure under scalar multiplication): Let $x \in W_1 \cap W_2$ and $\alpha \in \mathbb{F}$. Since $x \in W_1$, which is a subspace and thus closed under scalar multiplication, $\alpha x \in W_1$. Similarly, $\alpha x \in W_2$. Then $\alpha x \in W_1 \cap W_2$.
The set $W_1 \cap W_2$ satisfies all the properties required to be a subspace of V. □

P3. The following claim is **False**:

Claim. Let V be a vector space over $\mathbb{F}$ and suppose W_1, W_2, W_3 are subspaces of V such that $W_1 + W_3 = W_2 + W_3$. Then $W_1 = W2$.

Proof. (that the claim is false). Let V be the set $\mathbb{C}$ of complex numbers, let $W_1 = \mathbb{R}$ be the set of real numbers, let $W_2 = \mathbb{C} \setminus \mathbb{R}$ (purely imaginary numbers), and let $W_3 = \{ x + xi \mid x \in \mathbb{R} \}$ (complex numbers with equal real and imaginary parts). W_1, W_2, and W_3 are all subspaces of $\mathbb{C}$. The sum of sets $W_1 + W_2 = \mathbb{C}$, and $W_2 + W_3 = \mathbb{C}$, but certainly W_1 (the set of real numbers) does not equal the set of imaginary numbers W_2. □

Notes about the previous problem:
It's not difficult to see that $W_1 + W_2 = \mathbb{C}$: The subspace W_1 is the set of real numbers and W_2 is the set of imaginary numbers. Any $z \in \mathbb{C}$ is the sum of a real number and an imaginary number. To see why $W_2 + W_3 = \mathbb{C}$, let z be any complex number. Then $z = a + bi$ for some $a, b \in \mathbb{R}$. This can be re-written as

$$z = a + bi = (a + ai) + (b - a)i.$$

Clearly, $(b - a)i \in W_2$ (the imaginary numbers), and $(a + ai) \in W_3$.

In general, any line through the origin on a plane P that contains the origin is a subspace of P. For any two different lines X and Y lying on the same plane P and intersecting at the origin, their sum is that plane: $X + Y = P$. We can have any number of lines on P intersecting at the origin, all different, but the sum of any two of them is P.

P4. This claim is also **False**: Let V be a vector space over $\mathbb{F}$ and suppose W_1, W_2, W_3 are subspaces of V such that $W_1 \oplus W_3 = W_2 \oplus W_3$. Then $W_1 = W_2$.

Proof. The same counterexample from P3 works here, because the sums $W_1 + W_2$ and $W_2 + W_3$ are also direct sums, since $W_1 \cap W_2 = W_2 \cap W_3 = \{0\}$. □

Note: Prop 4.4.7 (part 2) states that $X \oplus Y$ is defined only if $X \cap Y = \{0\}$ (X and Y have only the 0 vector in common).